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THE FAILURE OF BRITTLE POROUS SOLIDS UNDER COMPRESSIVE STRESS STATES

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Abstract—When porous, brittle, solids are loaded in compression, small cracks grow from the pores. The extent of growth is calculated as a function of pore size and confining pressure, and compared with experiments on glass and on brittle plastics. The interaction of growing cracks with the surfaces of the sample and with each other is analysed in detail. This leads to the formulation of a theory of damage mechanics for brittle porous solids in multiaxial compression.

Résumé—Lorsqu'on soumet des solides poreux fragiles à une compression, de petites fissures se forment à partir des pores. Nous calculons leur croissance en fonction de la taille des pores et de la pression de confinement et nous la comparons avec des expériences sur des verres et sur des matières plastiques fragiles. Nous analysons en détail l'interaction de fissures en cours de croissance avec la surface de l'échantillon et avec les autres fissures. Ceci fermet de formuler une théorie de la mécanique de l'endommagement pour les solides poreux fragiles en compression multiaxiale.

Zusammenfassung—Werden poröse, spröde Festkörper im Druckversuch belastet, dann wachsen kleine Risse aus den Poren. Das Ausmaß des Wachstums wird in Abhängigkeit von Porengröße und Druck berechnet und mit Experimenten an Glas und spröden Polymeren verglichen. Die Wechselwirkung wachsender Risse mit der Probenoberfläche und untereinander wird ausführlich analysiert. Dies führt zur Formulierung einer Theorie des Schädigungsprozesses spröder poröser Festkörper unter multiachsialem Druck.

NOMENCLATURE

x, y, z	cartesian co-ordinate set, centred on hole centre
X, Y, Z	normalised co-ordinates defined by equation (A1)
$r, \theta, (\phi)$	cylindrical (spherical) co-ordinate set
E_0	Young's modulus of the uncracked body (MPa)
E	Young's modulus after cracking (MPa)
ν	Poissons ratio
$\sigma_1, \sigma_2, \sigma_3$	principal stresses (MPa)
λ	ratio of least compressive to most compressive principal stress, σ_3/σ_1
σ_{zz}, σ_{00}	local normal stress acting on crack surface (MPa)
K_I, K_I^B, K_I^1	mode I stress intensities (MPa $m^{1/2}$)
K_{IC}	fracture toughness (MPa $m^{1/2}$)
G	strain energy release rate (J/m^2)
σ	yield strength of the material (MPa)
r	radial distance from hole (m)
R	r/a
a	radius of hole or pore (m)
α	a constant; αa is the effective hole depth; $\alpha = 0.6$ (Fig. A1)
l	length of crack (m)
L	normalised crack length, $L = l/a$
b	plate thickness (m)
t	half-width of plate or column thickness (m)
I	second moment of area of a beam or column (m^4)
m, m_1, m_2	moment tending to bend column (Nm)
U	elastic energy (J)
T, S	axial and shear forces on a beam (N)
$F(L, \lambda)$	a function of L and λ
$G(X/L)$	a function of X and L
P	potential energy of cracked body, per crack (J)
ΔP	change in potential energy per crack (J)
V	sample volume (m^3)
N_A	number of holes per unit area (m^{-2})
f_A	the area-fraction of holes in a porous material, $f_A = \pi a^2 N_A$
C^*	dimensionless quantity defined by equation (20)

1. INTRODUCTION

Brittle solids fail in compression by a process of progressive microfracture. Acoustic emission shows that microfracturing begins at a stress of about one-half the ultimate failure stress, and that the number and length of the microcracks increases as the stress is increased further, causing the sample volume to increase even though the stress state is compressive. Near failure, the density of microcracks and the dilatency increases rapidly. Microscopy shows that individual microcracks propagate in a direction which is more-or-less parallel to the direction of the largest principal compressive stress until they link to form one of the several types of macroscopic failure.

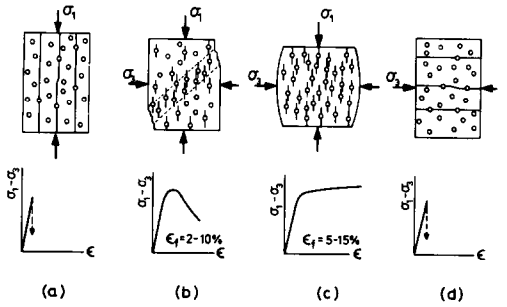


Fig. 1. The failure modes of porous brittle cylinders under an axial compression σ_1 as a function of confining pressure $\sigma_2 = \sigma_3$. Simple compression causes failure by "slabbing" shown at (a); modest confining pressures give the shear failure of (b); and large confining pressures give homogeneous microcracking, shown at (c).

The type of failure depends upon the confining pressure in the way shown schematically in Fig. 1. Unconfined samples, shown at (a) and (d), tend to fail by "vertical slabbing": the microcracks grow or combine to form contiguous vertical failure planes. At low and intermediate confining pressures, shown at (b), failure occurs in a shear band; the angle between this band and the compression axis varies with confining pressure, often following a simple Coulomb failure criterion. At high confining pressures, shown at (c), a single failure surface does not form; instead, the sample deforms in a pseudo-ductile mode with large-scale deformation being taken up by many, short, homogenously distributed microfractures.†

The individual microcracks initiate at flaws or stress concentrations within the solid: pores, or inclusions, or small (often grain-sized) cracks left by its earlier thermal or mechanical history. In this paper we analyse the growth of axial cracks from cylindrical holes in two dimensions, and from spherical pores in three dimensions; crack growth from a dispersion of small cracks is analysed in a companion paper [1]. The method is an approximate one, using simple analytical results of fracture mechanics and beam theory, checked against more precise numerical calculations when these are available. We further consider the interaction of a growing crack with the free surfaces of the sample containing it, and with other cracks. This leads to the formulation of a framework of damage mechanics which allows the stress–strain curve and the conditions for final macroscopic failure, to be computed. The theoretical results are compared with data from experiments on glass and PMMA plates and resin cylinders.

†We are concerned here with truly brittle solids (like glass and most silicates) only. Some materials (like marble and rock salt) are brittle in simple compression but become plastic under sufficiently large confining pressures.

It should be emphasised that the behaviour of brittle solids in compression is very complicated. This paper and its companion [1] do not answer all the questions raised by the observations; it is clear that much more work is required to develop reliable constitutive laws for compressive fracture. But there is evidence that the approach we develop here has considerable promise.

2. EXPERIMENTS: CRACK GROWTH FROM HOLES AND PORES IN GLASS AND BRITTLE PLASTICS

We studied crack growth from a circular hole by loading plates of glass and of PMMA in compression (Fig. 2). Figure 3 shows photographs of the way the cracks in PMMA plates extended stably as the compressive load (applied vertically) was increased.

The plates were 10 mm thick, 170 mm high and 50–100 mm wide. A central hole (and sometimes an array of holes) was drilled through each sample, with a diameter $2a$ of 5, 10 or 20 mm. Small starter cracks ($l \sim 0.02a$) were made at the top and bottom of each hole with a glass cutter (for the glass) or by tapping with a razor blade (for PMMA). The plates were loaded in axial compression at a strain rate of about 10^{-4}/s in a rigid jig which constrained the ends to prevent out-of-plane motion and rotation, and with edge-guides to suppress buckling [1]. The behaviour of the growing crack is extremely sensitive to the end conditions: misalignment, off-axis loading, and overconstraining the sample ends can all cause large systematic errors. Reliable results were obtained by loading through serrated hard-rubber pads, and by lining the buckling and bending guides with teflon shims. Crack growth was photographed and correlated with the load–deflection curve.

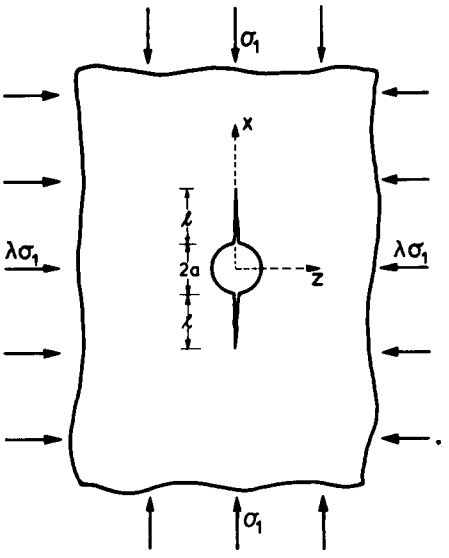


Fig. 2. Crack geometry and co-ordinate system for the 2-dimensional calculations.

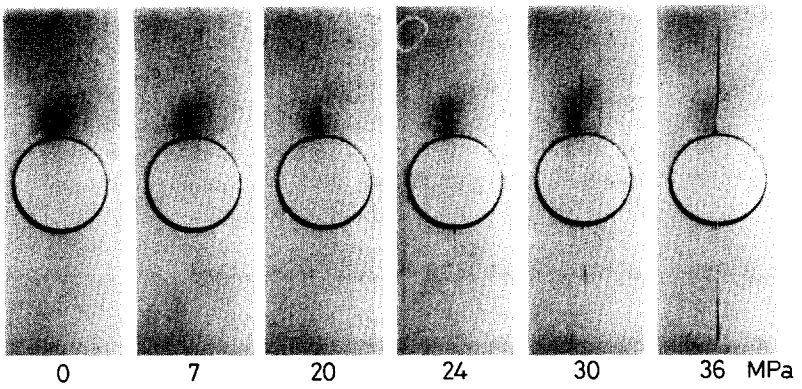


Fig. 3. PMMA plates with cylindrical holes, after loading in uniaxial compression. The cracks grow stably, requiring progressively larger loads to drive them further from the hole at which they nucleate.

Typical data for crack growth from a single hole are plotted in Figs 4 and 5, on normalised axes: the crack length l is plotted as l/a where a is the radius of the hole from which it grows; and the compressive stress σ_1 is plotted as $\sigma_1\sqrt{\pi a}/K_{IC}$ where K_{IC} is the fracture toughness of the material, using $K_{IC} = 0.6$ MPa $m^{1/2}$ for glass and 1.0 MPa $m^{1/2}$ for PMMA. (These are average values for slow crack growth [2].) Figure 4 shows crack growth data for glass. The cracks pop in to a length $l/a \approx 0.2$; thereafter growth is stable so that the stress rises steeply as the crack is driven further from the hole. Figure 5 shows data for PMMA: here, too, cracks pop in to $l/a \approx 0.2$ and then grow stably. Cracks grow more easily in narrow plates ($t/a = 5$) than in wide ones ($t/a = 10$).

Typical crack growth from an array of holes is shown in Fig. 6. The interaction causes cracks to nucleate earlier and grow further from the small holes than from the large ones. When the cracks are large they deviate towards each other and finally link. The data are plotted in Fig. 7.

Studies of crack growth in 3 dimensions are more difficult. We examined the nucleation and growth of cracks from spherical pores embedded in a block of epoxy, loaded in simple compression. Figure 8 shows a typical sample.

To make the specimens, a 1 cm diameter sphere of expanded polystyrene was suspended by a diagonal thread at the centre of a cubic mould which was then filled with transparent epoxy resin to which hardener had been added. When the epoxy had cured, the cube was removed from the mould and machined into a right circular cylinder with a diameter of 64 mm and a height of 100 mm. The cylinders were loaded in compression at a strain rate of about 10^{-4}/s in a

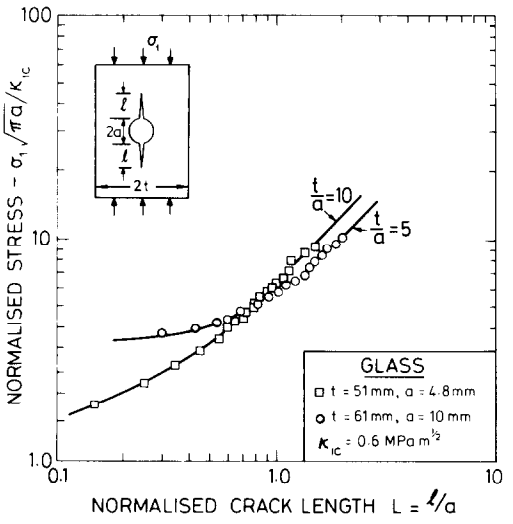


Fig. 4. Typical data for crack growth in glass plates in simple compression. The initial stages of growth show much variation; the later stages showed a systematic dependence of crack length on the ratio t/a of plate width to hole radius.

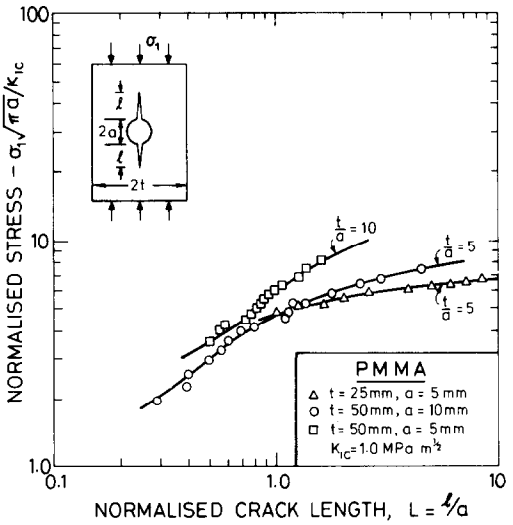


Fig. 5. Typical data for crack growth in PMMA. Like glass, the initial stages of growth show much variation; the later stages showed a systematic dependence on t/a.

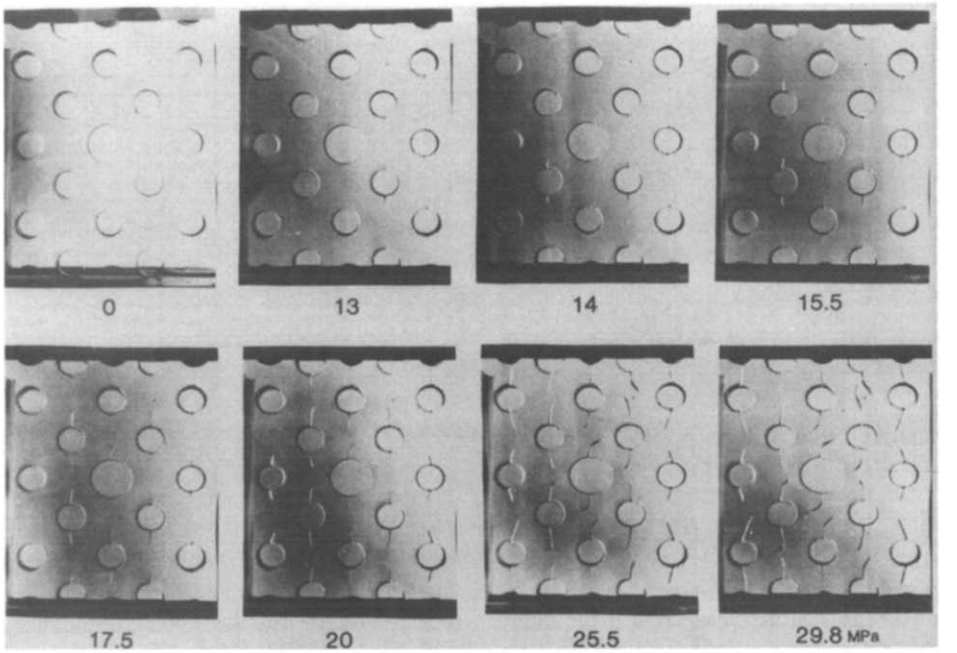


Fig. 6. Crack growth from an array of holes in PMMA.

compression jig with teflon shims between the sample ends and the plattens. The cracks shown in Fig. 8 initiated suddenly with an audible pop and then grew stably as the load was increased further. They lie parallel to the compression axis, with lengths l/a ranging from 0.8 to 2. Taking $K_{IC} = 0.5$ MPa m$^{1/2}$ (measured independently for this epoxy), the normalized stress ($\sigma_1\sqrt{\pi a}/K_{IC}$) is 58 for this sample.

3. MODELLING OF CRACK GROWTH FROM A CYLINDRICAL HOLE IN A BRITTLE PLATE UNDER COMPRESSION

In this section we analyse the growth of a crack from a cylindrical hole of radius a, contained in a plate subjected to a remote biaxial stress field (σ_1, σ_3), as shown in Fig. 2. The hole in an infinite plate is considered first; then the interaction of the growing crack with the edges of a finite plate is analysed; and finally the influence of plasticity is discussed. The physical basis and the results of the analysis are described here; the mathematical details are contained in Appendix 1.

3.1. Crack growth from a hole in an infinite plate

Consider an elastic plate containing a hole of radius a and subjected to principal stresses σ_1 and σ_3. The stresses are treated as positive when tensile, negative when compressive; then σ_1 is the most negative (most compressive) and $\sigma_3 = \lambda\,\sigma_1$ is the most positive (least compressive) principal stress (Fig. 2). The stresses are concentrated by the hole such that, for a simple compression, the stress at the north and south poles of the hole is *tensile* and equal in magnitude to $|\sigma_1|$. Figure 9 (broken line) shows how this tensile stress (normalised by σ_1) falls rapidly with distance $X = x/a$ (measured radially outward from the surface of the hole surface), becoming compressive when $X \geq 2/3$. A confining stress changes the local stresses: when σ_3 is compressive also, the local stress is less tensile; and if $\sigma_3/\sigma_1 > 1/3$, no tensile stress appears anywhere on the surface of the hole.

When local tensile stresses exist, they are always greatest at the north and south poles. A pair of

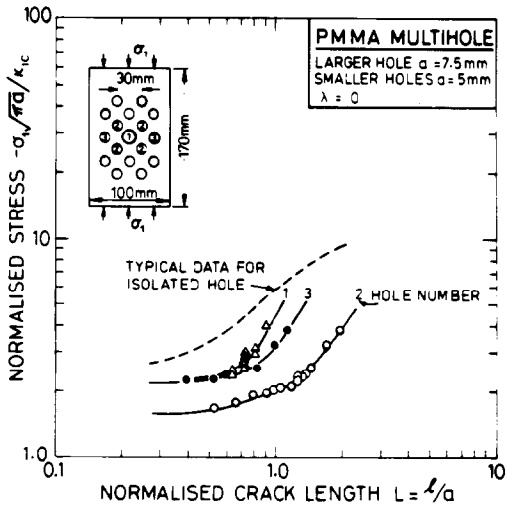


Fig. 7. Typical data for crack growth from an array of holes in PMMA. The cracks grow more easily when another crack is nearby.

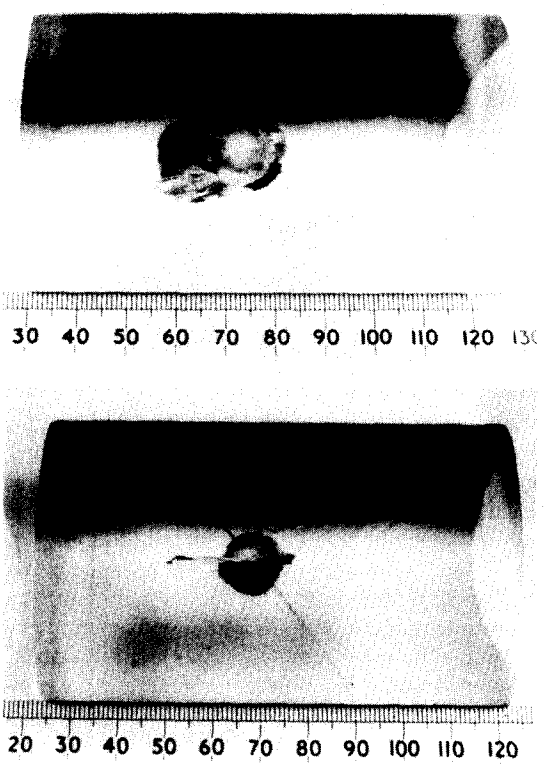


Fig. 8. A resin cylinder containing a spherical hole from which compression cracks have grown.

cracks of length l emanating from these points is subject to mode 1 loading, and a stress intensity K_I is associated with each. Standard methods are available to calculate K_I: they involve an appropriate integral of the local stress field over the surface of the crack. Approximate ways of doing this are described in Appendix 1, and compared with published solutions to an analogous problem. The result has the form [5]

$$K_I = -F(\lambda, L)\sigma_1\sqrt{\pi l} \tag{1}$$

where l is the crack length (measured from the edge of the hole), $L = l/a$ where a is the hole radius, and $\lambda = \sigma_3/\sigma_1$. The function $F(\lambda, L)$ is tabulated in the Appendix and plotted in Fig. 10. The symbols and lines refer to the different ways of estimating K_I: for practical purposes they are in complete agreement.

It is useful to have an analytical expression to describe $F(\lambda, L)$. The results are well fitted by the equation (suggested by the equations from which it is derived in Appendix 1)

$$F(\lambda, L) \simeq \frac{1.1\,(1 - 2.1\,\lambda)}{(1 + L)^{3.3}} - \lambda \tag{2}$$

from which

$$K_I = -L^{1/2}\left\{\frac{1.1\,(1 - 2.1\,\lambda)}{(1 + L)^{3.3}} - \lambda\right\}\sigma_1\sqrt{\pi a}. \tag{3}$$

when $\lambda = 0$ (no confining stress), $F(\lambda, L)$ and thus K_I are finite for all L; but when $\lambda > 0$, both go to zero at a finite distance from the hole. Cracks cannot grow longer than the value of L for which $F(\lambda, L) = 0$. In the limit $\lambda < 0$ and $L \gg 1$, K_I reduces to $K_I = \sigma_3\sqrt{\pi l}$, the obvious result for a crack of length l in a simple tensile field.

The cracks grow until the stress intensity K_I becomes equal to the fracture toughness K_{IC}. Then the stable crack length, L, is defined by

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = -\frac{1}{L^{1/2}\left\{\dfrac{1.1\,(1 - 2.1\,\lambda)}{(1 + L)^{3.3}} - \lambda\right\}}. \tag{4}$$

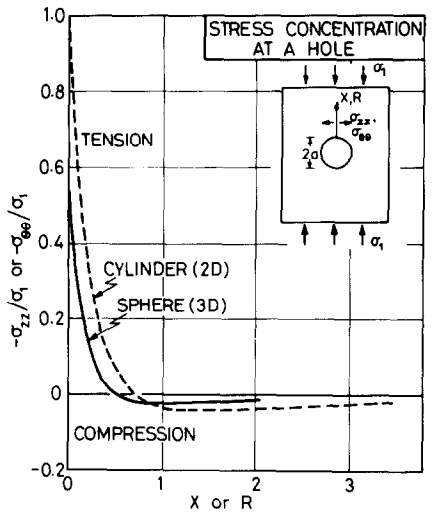


Fig. 9. The stress $\sigma_{zz}(X, 0)$ and $\sigma_{\theta\theta}(R, 0)$ along the pole axis of the cylindrical and the spherical pore, under uniaxial compression σ_1.

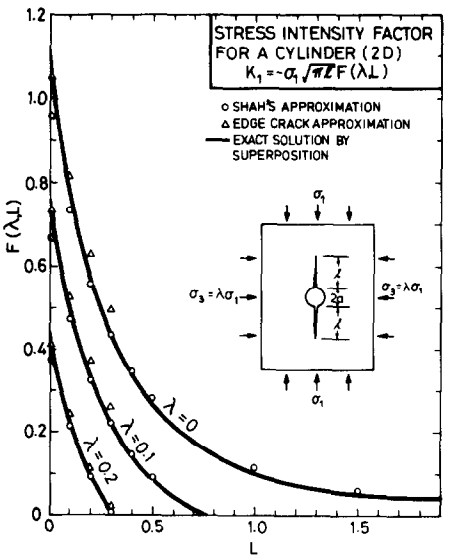


Fig. 10. The integral $F(\lambda, L)$ for the cylindrical hole. The points are calculated from the approximate equations given in Appendix 1; the lines obtained by superposition of the exact numerical calculations of Sih [5].

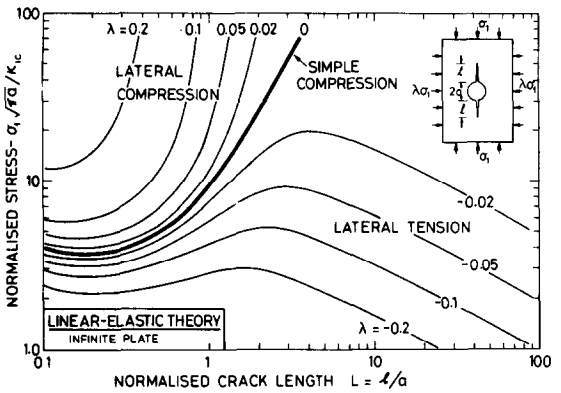


Fig. 11. Crack extension, l/a, as a function of stress in an infinite plate [equation (4)]. When σ_3 is zero or compressive (λ positive), crack growth is stable (after initial pop-in to $l/a = 0.2$). When σ_3 is tensile (λ negative) cracks ultimately become unstable.

This result is plotted in Fig. 11, which shows $-\sigma_1\sqrt{\pi a}/K_{IC}$ as a function of L, for various values of λ. The stress required to initiate a crack from a perfectly smooth hole is large; in practice, cracks must initiate from flaws on the surface of the hole. For flaws of length less than $L \approx 0.1$ the growth is at first unstable (as we observed in our experiments). Thereafter, the crack grows in a stable way, requiring an increase in stress for each increment of crack growth. A confining pressure $\sigma_3 = \lambda\sigma_1$ tends to suppress crack growth; when $\lambda \geq 1/3$ cracks are suppressed completely. When $\lambda = 0$ (simple compression) equation (4) becomes

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = -\frac{(1+L)^{3.3}}{1.1\sqrt{L}}. \tag{5}$$

The stress at first falls as $1/\sqrt{L}$, passes through a minimum at $L = 0.18$ and then rises again, ultimately (for long cracks) as $L^{2.8}$. When λ is negative (so that σ_3 is tensile) the cracks at first grow stably but ultimately become unstable and grow without limit.

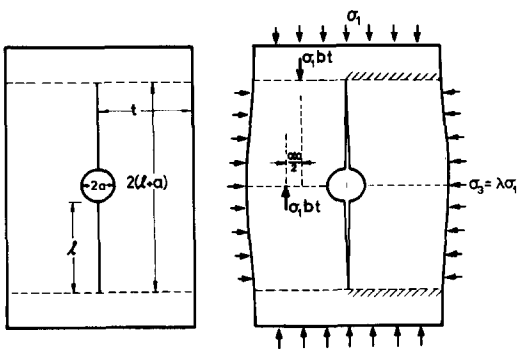


Fig. 12. The bending displacements in a finite plate containing a hole and associated cracks.

3.2. *Interaction of the cracks with the surfaces*

The experiments of Section 2 are compared with the predictions of the theory at the end of this section. The comparison shows that the cracks grow in the way described by equation (4) (and plotted in Fig. 11) when they are short and the plate containing them is large, but that they extended much further than this when their length l becomes comparable with the width w of the plate. We have studied cracks at holes in silicone rubber plates (which allow large elastic deflections) and made detailed measurements of the shape change of the PMMA plates during loading. The observations and measurements all show that *bending displacements* appear in plates when the crack length becomes comparable with the plate width. These bending displacements (documented in slightly more detail in Ref. [1]) are shown schematically in Fig. 12: the cracks split the plate into two columns or beams, built-in at the ends and of length $2(l+a)$, which bend outwards as the plate is loaded; the bending moment is caused by the change of section at the centre of each beam. It is analysed in Appendix 1 where it is shown that the axial compression σ_1 causes an outward bending of the beam; it is counteracted by the confining stress σ_3 which acts as a distributed load on the outer surface of each beam, bending to bend it back in again. The bending gives an additional contribution to the stress intensity.

It is evaluated by standard methods. The elastic energy associated with the bending is first calculated and differentiated with respect to crack length l to give the crack driving force G; K_I is then obtained from the relationship $K_I = \sqrt{EG}$. This is done in Appendix I with the result (simplified but without important loss of accuracy)

$$K_I^B = -\left(\frac{3}{\pi}\right)^{1/2}\left(\frac{a}{t}\right)^{1/2}\frac{\alpha^2}{(L+1)}\sigma_1\sqrt{\pi a}\left\{\frac{1-\frac{2(L+1)^3}{3}\frac{a}{\alpha^2}\frac{\lambda}{t}}{1+\frac{12}{\pi^2}\frac{\sigma_1}{E_0}\frac{a^2}{t^2}(L+1)^2}\right\} \tag{6}$$

where $t = w/2$, α is a constant equal to 0.6 and E_0 is Young's modulus for the material. This expression describes the interaction of the crack with the surfaces of the sample in a multiaxial stress field. It consists of three parts. The part in front of the curly brackets is the contribution to K_I from the outward bending caused by σ_1. The term above the horizontal line in the curly brackets describes the restraining effect of the confining stress σ_3. Finally, the term below the horizontal line is a factor which corrects for the change in geometry caused by beam bending: it goes to zero (causing K_I^B to go to infinity) at the Euler buckling load of the column. Note that K_I^B is negligible for wide plates ($t/a \gg 1$), but becomes increasingly important as t/a approaches 1.

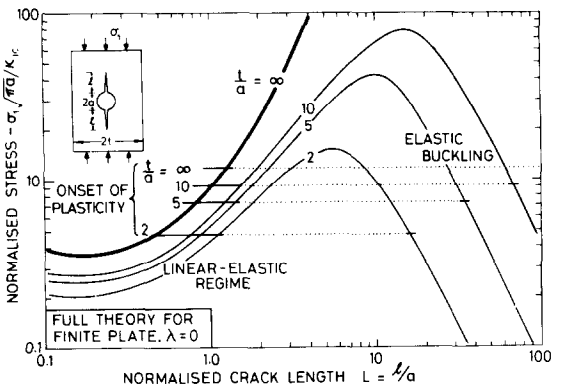


Fig. 13. Crack extension l/a as a function of stress for a hole in a finite plate [using the sum of equation (3) and (6)]. Glass and other truly brittle materials should follow the full lines; for PMMA and other elastic-plastic materials, the elastic solution may be truncated by plasticity [broken lines, equation (8)].

The total K_I driving crack growth is given by adding equation (6) to (3). As before, the crack grows until K_I falls to K_{IC}. The resulting crack length is plotted as a series of full lines, for various t/a, in Fig. 13.

There is one further complication. Some materials (glass, for instance) are elastic under all loading conditions. Others (like PMMA) are linear elastic at low stresses but yield plastically at higher stresses. We have carried out tests on both glass and PMMA samples (Section 2) and find that plasticity can play a role in the compression cracking of the PMMA. When a beam of thickness t and depth b is subjected to an axial stress σ_1 and a bending moment m, it starts to yield when the maximum surface stress reaches the yield strength, that is, when

$$\sigma_1 + \frac{6m}{bt^2} = \sigma_y.$$

The beams which make up the compression sample of Fig. 12 are narrowest at the hole. It is here that both the axial stress and the bending moment ($m = \sigma_1 bt\alpha a/2$) are largest. Yielding starts here when

$$|\sigma_1| = \frac{\sigma_y}{1 + 3\alpha a/t} \tag{7}$$

and the effect of this is to wedge open the crack yet further, greatly increasing K_I at its tip. This is shown in Fig. 13, which illustrates that the elastic solution (full lines) will be truncated when equation (7) is satisfied, that is, when

$$-\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{\sigma_y\sqrt{\pi a}}{K_{IC}(1 + 3\alpha a/t)}. \tag{8}$$

3.3. Comparison of theory with experiment for holes in plates

The data for crack-growth in glass are compared with the elastic theory in Fig. 14. (The modulus of glass, 69 GPa, is so large that the elastic buckling lines which appear in Fig. 12 all lie off this diagram to the right.) The data lie close to the theory once the cracks have grown to a length l/a of more than 0.5; and the theory appears to explain the observed dependence of crack growth on plate thickness (measured by t/a).

Data for PMMA are compared with theory in Fig. 15. Buckling appears because the modulus of PMMA (3.5 GPa) is much lower than that of glass. The elastic theory is truncated by plasticity [equation (8)] using our measured value of yield strength of PMMA (94 MPa). The combined elastic–plastic analysis explains the observed dependence of crack growth on plate thickness (t/a) and gives some understanding of the plateau in the growth curve for cracks in PMMA, a feature which is absent in the data for glass.

4. MODELLING OF CRACK GROWTH FROM A SPHERICAL PORE IN A BRITTLE SOLID UNDER COMPRESSION

Real materials (rock and minerals, for instance) contain spherical pores which are totally enclosed, from which cracks may grow when the material is loaded in compression. In this section we describe

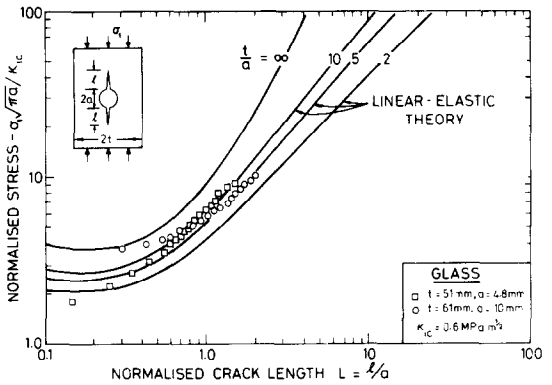


Fig. 14. Comparison of the theory with data for glass.

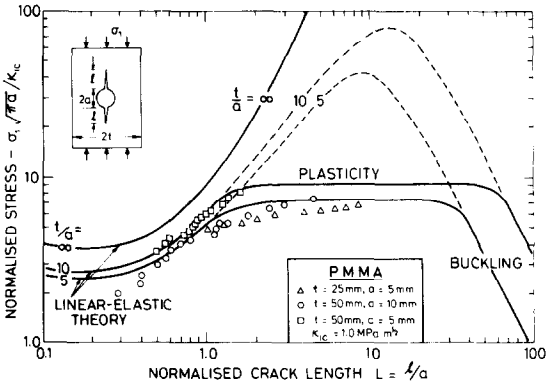


Fig. 15. Comparison of the theory with data for PMMA.

an approximate analysis of crack growth from a single, enclosed, pore, including the interaction of the crack with external surfaces and the final buckling instability which leads to catastrophic growth. The methods are approximate (though they are dimensionally sound), and are based on simple adaptations of the analysis of cracking from a circular hole in a plate. Details, where necessary, are given in Appendix 2.

4.1. Crack growth from pore in an infinite body

Consider an elastic body containing a spherical pore, subjected to principal stresses σ_1, σ_2 and σ_3 (Fig. 16). The stresses are positive when tensile, and σ_1 is the most compressive and $\sigma_3 = \lambda\sigma_1$ the least compressive principal stress. We focus on cracks which grow in the plane normal to the σ_1, σ_3 plane and assume that σ_2 has no influence. The stresses are concentrated by the pore [3] such that, in simple compression, the stress $\sigma_{\theta\theta}$ at its north and south poles is tensile, and equal in magnitude to

$$\sigma^0_{\theta\theta} = -\frac{3(1+5\nu)}{2(7-5\nu)}\sigma_1 \approx -\frac{1}{2}\sigma_1$$

where ν is Poisson's ratio. The stress concentration is less than for a hole in a plate, and falls off faster with distance from the edge of the hole as shown in Fig. 3 (full line). The equation for $\sigma_{\theta\theta}$ as a function of distance $R = r/a$ is given in Appendix 2, where it is used to calculate the stress intensity K_I which we write (as before) in the form

$$K_I = -F(\lambda, L)\sigma_1\sqrt{\pi l}$$

where l is the current length of the crack and $L = l/a$ (Fig. 16). The results of the calculations, for $\nu = 0.25$, are plotted as $F(\lambda, L)$ in Fig. 17, and listed in Table 2. When there is no confining stress ($\lambda = 0$), $F(\lambda, L)$ and thus K_I are positive for all $L = l/a$. A small confining stress ($\lambda = 0.05$, for instance) causes $F(\lambda, L)$ to fall to zero at a finite crack length: cracks cannot grow beyond the point at which $F(\lambda, L) = 0$.

It is useful to have an analytical expression for $F(\lambda, L)$. Figure 17 shows that the results are well fitted by

$$F(\lambda, L) = \frac{0.62(1-1.8\lambda)}{(1+L)^{4.1}} - \lambda \qquad (9)$$

from which

$$K_I = -L^{1/2}\left\{\frac{0.62(1-1.8\lambda)}{(1+L)^{4.1}} - \lambda\right\}\sigma_1\sqrt{\pi a} \qquad (10)$$

In the limit $\lambda < 0$ and $L \gg 1$, K_I reduces to $K_I = \sigma_3\sqrt{\pi l}$, the obvious result for a crack of length l in a simple tensile field.

The cracks grow until K_I falls to K_{IC}. The stress state which will drive the cracks to a length L is defined by

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{-(1+L)^{4.1}}{[0.62(1-1.8\lambda) - \lambda(1+L)^{4.1}]\sqrt{L}}. \qquad (11)$$

As with the cylindrical hole, a value of λ exists above which there is no tensile stress at any point on the hole surface, and no crack can form. For $\nu = 0.25$, the result is $\lambda = 0.29$; for $\nu = \frac{1}{3}$ it is 0.31. When $\lambda = 0$ (simple compression) we find

$$-\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{(1+L)^{4.1}}{0.62\sqrt{L}}. \qquad (12)$$

The stress at first falls as $1/\sqrt{L}$, passes through a minimum at $L = 0.14$ and then rises again, ultimately as $L^{3.6}$. The stress required to nucleate a crack from a perfectly smooth hole is large, so cracks must nucleate from flaws on the hole surface. Cracks of length $L < 0.14$ grow unstably; cracks which are longer than this grow in a stable way.

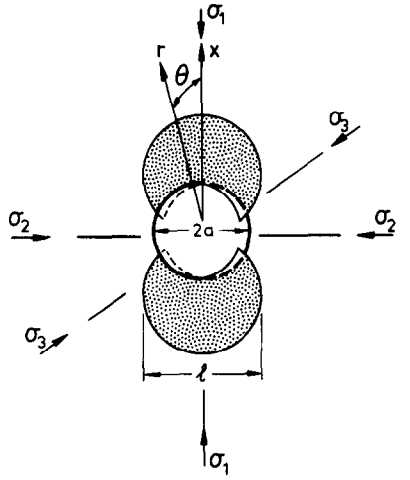


Fig. 16. Crack growth from an enclosed spherical pore, showing the co-ordinate system.

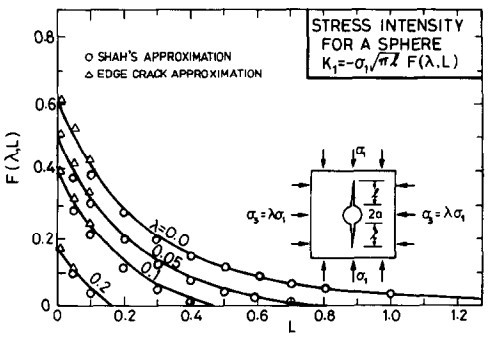


Fig. 17. The integral $F(L, \lambda)$ for a spherical pore. The points are calculated by the method described in the Appendix; the full lines show the approximate equation (9) given in the text.

4.2. Interaction of the cracks with the surfaces

As in the 2-dimensional problem of Section 3, the growing cracks interact with nearly free surfaces by mechanisms which involve bending and buckling. Both processes increase the crack opening, and thus K_I.

A full analysis for either process in the 3-dimensional case is obviously difficult. We adopt, instead, a scaling procedure which allows the 2-dimensional results to be adapted appropriately. Imagine that Fig. 12 now shows a section through a sample containing a spherical pore from which contained, semicircular cracks emanate at the north and south poles, as in Fig. 16. The cracks extend parallel to two of the free surfaces of the specimen. The ligaments between the cracks and surface can be thought of as rectangular plates of height $2(l+a)$ and width $2l$, built-in around their perifery. Because they are built-in in this way, their bending stiffness and their resistance to buckling are both greater than those of the beams or columns analysed in Section 3. But if the new bending stiffness and buckling load can be calculated, then the 2-dimensional result [equation (6)] can be modified in a simple way to describe the 3-dimensional problem.

The bending stiffness S_p of a rectangular plate of height h, width b and thickness t, built-in along all four edges, is known [6]. It is larger than that, S_B, of a beam of the same dimensions, built-in top and bottom, by a factor which depends on the ratio h/b. For $h/b = 2(l+a)/l \approx 2$, the factor is

$$\frac{S_p}{S_B} = 18.$$

The buckling load $(P_c)_p$ of the same built-in plate is also known [7]. It, too, is greater than that of a beam or column of the same dimensions, built-in top and bottom, $(P_c)_B$, by a factor which, again, depends on the ratio h/b. For $h/b \approx 2$ the factor is

$$\frac{(P_c)_p}{(P_c)_B} = 15.$$

Roughly speaking, then, the plate is about 18 times stiffer, and has roughly 15 times the buckling load, of the equivalent column. Then (if the calculation of Section 3 is repeated) the additional stress intensity caused by interaction with surfaces [equation (6)] becomes

$$K_I^B = -\frac{1}{18}\left(\frac{3}{\pi}\right)^{1/2}\left(\frac{a}{t}\right)^{1/2}\frac{\alpha^2}{(L+1)}\sigma_1\sqrt{\pi a}\left\{\frac{1-\frac{2}{3}\frac{(L+\alpha)^3 a}{\alpha^2 t}\lambda}{1+\frac{4}{5\pi^2}\frac{a^2}{t^2}(L+1)^2\frac{\sigma_1}{E_0}}\right\} \quad (13)$$

where $t = w/2$. The origin of the terms is described in Section 3. The general pattern of behaviour resembles that in two dimensions, but requires loads which are considerably greater.

The total K_I driving crack growth is given by adding equation (10) and equation (13); as always, the crack grows until K_I falls to K_{IC}.

4.3. Comparison of theory with experiment for pores in 3-dimensional bodies

The way in which crack length depends on axial stress is plotted, for various stress states λ, in Fig. 18. [It is plotted by summing equation (10) and (13) for K_I, equating the sum to K_{IC} and solving for $\sigma_1\sqrt{\pi a}/K_{IC}$.] The behaviour looks like that for the 2-dimensional case (Fig. 13) displaced to higher stresses. As before, the crack pops-in to a length L of about 0.2, and thereafter, if all three principal stresses are compressive ($\lambda \geq 0$) it grows stably. If one principal stress is tensile ($\lambda > 0$) the crack eventually becomes unstable and runs without limit.

Our measurements for crack extension in simple compression are plotted as a bar on the figure (our cracks popped-in to a depth of about $L = 1$ because we could not create starter cracks to initiate them in a more controlled way). All that can be said is that the theory appears to be consistent with the very limited data.

5. THE DAMAGE-MECHANICS OF POROUS BRITTLE SOLIDS

In the previous sections, we analysed the growth of a crack from an isolated hole. We now ask: how does an array of such cracks interact? And what is the criterion for final failure? Our ultimate goals are to predict the *failure surface* (the combination of σ_1, σ_2 and σ_3 which cause final failure), and to explain the *progression of fracture modes* from slabbing to shear failure to pseudo-ductility as the pressure is increased (Fig. 1).

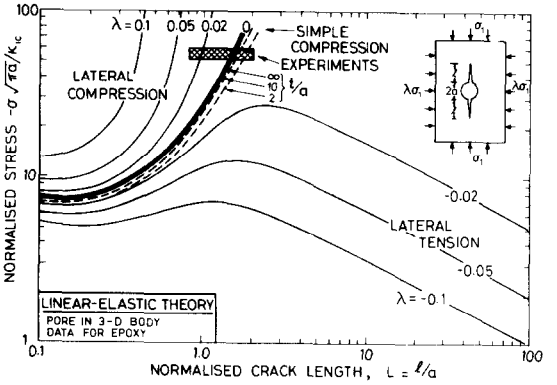


Fig. 18. Crack extension l/a as a function of stress for a pore in a finite body [using equation (10) and (13)]. Data from pores in epoxy resin ($K_{IC} = 0.5$ MPa $m^{1/2}$) are shown. Plasticity will truncate the elastic theory, as in Fig. 13.

5.1. The idea of a unit cell and a unit beam

Consider a body containing a random array of spherical holes, N_A of them per unit area, as shown in Fig. 19(a). The array can be typified by a regular grouping with a spacing equal to the average (roughly $1/\sqrt{N_A}$) in the array, as shown in the second column. The structure is then characterised by the shaded square unit cell of side t. There are two such cells per hole, and they pack to fill space completely. The diagonals of the cells can be thought of as a network of connected struts (in two dimensions) or of plates (in three). It is sometimes useful to think of the forces which are applied to the material as transmitted through this network as a system of axial forces, shear forces, and bending moments. The statics and kinematics of the structure can then be analysed approximately in terms of the forces acting on a single, typical, unit cell.

When cracks emanate from the holes, the shape of the unit cell changes. Figure 19(b) and (c) shows how the cracks grow between each other, carving the solid into beams which become narrower and longer, as the cracks extend. (The position of the holes is identical in the three parts of the figure.) Cracks which were not neighbours when they were short become neighbours as they grow longer—although the total number of cracks has not changed (it is always $2N_A$ per unit area). Because of this, the unit cell associated with a crack has constant area, it just becomes longer and thinner as shown in the second column of Fig. 19. If we identify the cell length with the crack length $(l + a)$, then the cell width t is given by

$$t(l + a) = \frac{1}{2N_A}. \tag{14}$$

Figure 19 shows how, as the cracks grow longer, these ligaments are increasingly well approximated by a network of struts alternately at angles $\pm\theta$ to the vertical (σ_1) direction, where

$$\tan\theta = \frac{a}{l + a}. \tag{15}$$

The length of a typical beam is

$$l' = \frac{l + a}{\cos\theta} \tag{16}$$

and, if all the material is to be accounted for, the average beam thickness, t', is again given by

$$l't' = \frac{1}{2N_A} \quad \text{or} \quad t' = t\cos\theta. \tag{17}$$

The forces applied to the material are transmitted through the strut-like ligaments between the cracks

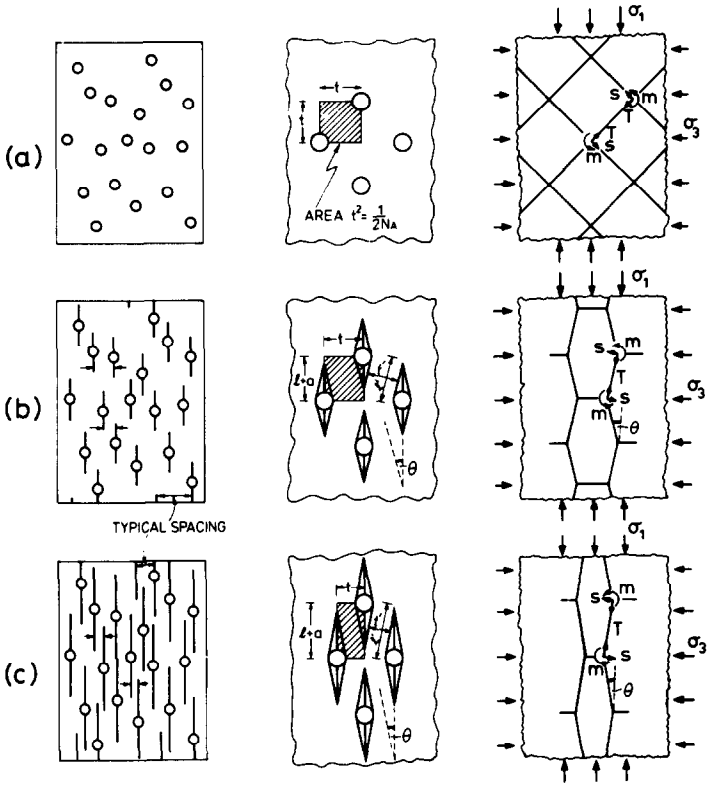


Fig. 19. The unit cell and beam. As cracks extend, the body is divided into ligaments which can be thought of as a network of beams through which the remotely applied stresses are transmitted as a set of axial forces and bending moments.

(Fig. 19, last column). As the cracks grow the beams become longer and more slender, and they rotate towards the vertical. Because of this the forces acting on them, and the elastic energy they contain, changes as the cracks grow. The derivative of this energy with respect to crack length gives the crack driving force G, from which the stress intensity K_1 at the crack tip can be calculated. The method (because it deals with the ligaments between cracks) automatically includes crack-crack interaction.

5.2. The stress intensity and crack extension

When the cracks are short, the unit-strut is short and fat; the bending deflection is negligible. Then the cracks do not interact and the stress intensity at the crack tip is that of an isolated crack; it is given by equation (3). But as the cracks grow, the unit strut becomes longer and thinner and the moments exerted on it by the remote stresses σ_1 and σ_3 cause bending deflections, and the cracks interact. This interaction is analysed in Appendix 3, using the method described above, with the result that bending gives an additional contribution to the stress intensity, K_1', of

$$K_1' = -\frac{\sqrt{2}}{\pi} f_A^{1/2}(L+1)^{1/2}\sigma_1\sqrt{\pi a} \times \left\{\left[1-\frac{8}{\pi}f_A\lambda(L+1)^3\right]\left[1-\frac{2}{\pi}f_A\lambda(L+1)^3\right]\right\}^{1/2}. \quad (18)$$

The term outside the curly brackets is the crack-opening stress intensity caused by the major compression σ_1; the term inside describes the crack-closing intensity caused by the confining stress σ_3. (If the cracks grow long enough, the buckling correction, described in Section 3, should be added: for simplicity we shall omit it here.) The total stress intensity is given by the sum of this and equation (3). As always, the cracks grow until K_1 falls to K_{1C}. The result is plotted in the upper part of Fig. 20, which shows how, in simple compression, the cracks grow stably at first, and then interact (when $L \approx 2$) in a way which makes them unstable [the instability is caused by the term $(L+1)^{1/2}$ in equation (18)].

5.3. The modulus of the cracked porous solid

As cracks grow from the holes, the modulus of the solid falls. Since the cracks are aligned, it also becomes anisotropic: the modulus in the principal compression direction differs from that in the other two. We now calculate the modulus in this direction, and use it to derive the stress–strain curve for the material.

Consider a sample containing a population of holes from which cracks extend (as in Fig. 19). Temporarily seal up the cracks and apply the compressive stress σ_1: the potential energy P of the sample and its loads is $-\frac{1}{2}\sigma_1^2 V/E_0$ where V is the volume of the sample and E_0 is the modulus of the sample with no cracks. Now, without changing loads, release the cracks. There is a further change in the potential energy of the system, of ΔP per hole. The effective Young's modulus E is defined by

$$\frac{1}{2}\frac{\sigma_1^2}{E}V = \frac{1}{2}\frac{\sigma_1^2}{E_0}V + \frac{N_A V}{b}\Delta P \quad (19)$$

where N_A is the number of holes per unit area and $N_A V/b$ is the number of holes in the sample.

For dimensional reasons, ΔP must take the form

$$\Delta P = \frac{C^*\sigma_1^2 a^2 b}{E_0} \quad (20)$$

where C^* is a function of L and λ, to be determined. Inserting this into equation (19) gives

$$E = E_0\left\{\frac{1}{1+2C^*N_A a^2}\right\}. \quad (21)$$

The problem reduces to that of calculating C^*.

Consider non-interacting cracks first. The change in potential energy caused by the extension of the cracks *without* interaction, can be calculated from the expression for K_1. For the two-dimensional case K_1 is well described by equation (3); then

$$\Delta P = 2b\int_0^l \frac{K_1^2}{E_0}\,dl = \frac{2b\sigma_1^2\pi}{E_0} \times \int_0^l l\left\{\frac{1.1(1-2.1\lambda)}{(1+L)^{3.3}} - \lambda\right\}^2 dl. \quad (22)$$

The assumption that cracks do not interact implies that L is small; then $1+L \approx 1$, and

$$\Delta P \approx \frac{\sigma_1^2\pi a^2 L^2 b}{E_0}(1-3\lambda)^2$$

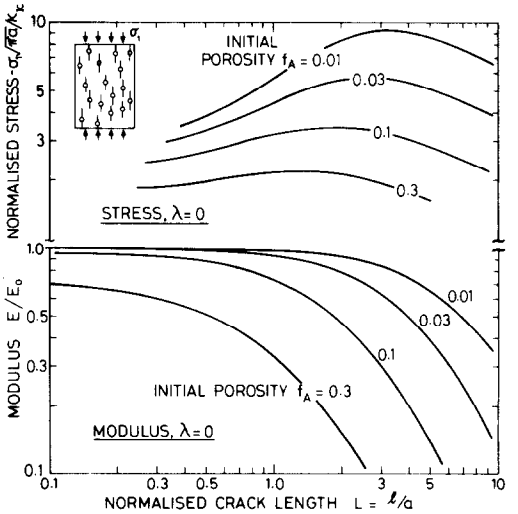


Fig. 20. The change in axial stress σ_1 and modulus E as cracks grow and interact. The ratio of the stress (upper diagram) to the modulus (lower diagram) at a given crack length gives the strain.

from which [by comparison with equation (20)]

$$C_1^* = \pi L^2 (1 - 3\lambda)^2 \tag{23}$$

for non-interacting cracks.

When cracks interact, the ligaments between them bend as illustrated in Fig. 19. This is analysed in Appendix 3, where we show that there is then an additional change in the potential energy of the system; it is [equation (C9)]

$$\Delta P = \frac{\sigma_1^2 b}{E_0} a^2 N_A (l + a)^2 \left\{1 - \frac{2(l + a)^3 N_A \lambda}{a}\right\}^2.$$

Again, comparing this with equation (20) shows that there is a second contribution to the drop in modulus, characterised by

$$C_2^* = N_A a^2 (L + 1)^2 \{1 - 2N_A a^2 (L + 1)^3 \lambda\}^2. \tag{24}$$

Substituting $C^* = C_1^* + C_2^*$ into equation (21) gives an equation relating modulus to the density and length of the cracks

$$E = \frac{E_0}{1 + 2\pi N_A a^2 L^2 (1 - 3\lambda)^2 + 2N_A^2 a^4 (L + 1)^2 \times (1 - 2N_A a^2 (L + 1)^3 \lambda)^2} \tag{25}$$

The modulus is plotted against crack length in the lower part of Fig. 20, which shows how E decreases as the cracks grow.

5.4. *Damage, and damage growth*

It is convenient to define the *crack damage* by

$$D = \pi l^2 N_A = \pi a^2 L^2 N_A.$$

It is dimensionless, and measures (roughly) the area influenced by cracks, per unit area. The modulus can now be expressed in terms of the damage, D, the stress ratio λ and the porosity f_A, where

$$f_A = \pi a^2 N_A.$$

The result is

$$E = \frac{E_0}{1 + 2D(1 - 3\lambda)^2 + \frac{2}{\pi^2} f_A^2 \left(\left[\frac{D}{f_A}\right]^{1/2} + 1\right)^2 \times \left\{1 - \frac{2}{\pi}\left[\left(\frac{D}{f_A}\right)^{1/2} + 1\right]^3 \lambda\right\}^2} \tag{26}$$

5.5. *The stress–strain curve*

When an axial compression σ_1 is applied to a porous sample in which the current crack length is l (and damage is D) the axial strain is given by

$$\varepsilon = \frac{\sigma_1}{E}. \tag{27}$$

As the cracks grow the modulus decreases and the strain becomes larger. But each value of stress σ_1 is associated [via equation (3) plus (18)] with a certain crack length, and thus damage. The stress–strain curve is calculated by treating crack length (or, equivalently, damage) as the independent parameter; at each value of L (or D) a value of σ_1 is found from equation (3) plus (18), and a value of E from equation (25). Strain is calculated from equation (27). It is convenient to plot stress in the dimensionless units used earlier

$$\tilde{\sigma} = \frac{\sigma_1 \sqrt{\pi a}}{K_{IC}}$$

and strain in corresponding units

$$\tilde{\varepsilon} = \frac{\sigma_1}{E} \sqrt{\pi a} \frac{E_0}{K_{IC}}.$$

Figure 20 showed how the stress and the modulus depended on crack length. The strain is the ratio of these two, at each crack length. Stress–strain curves in simple compression for different initial porosities are shown in Fig. 21: the stress rises, passes through a maximum (the failure stress) and then decreases; the peak is the failure strength of the material. The effect of confining pressure on crack growth is shown in Fig. 22: as the cracks grow the stress rises through

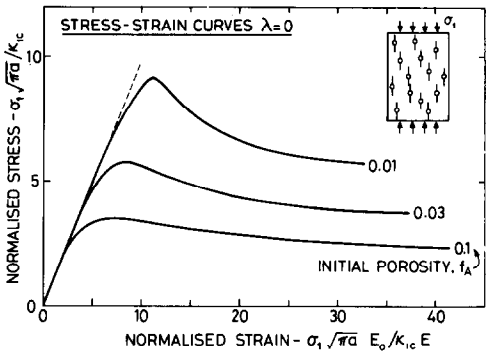


Fig. 21. Stress–strain curves in simple compression for a cracked, porous solid. As the cracks propagate further from the holes and interact, the stress at first rises and then falls and the modulus decreases; the result is a peaked stress–strain curve. The peak is the macroscopic failure strength.

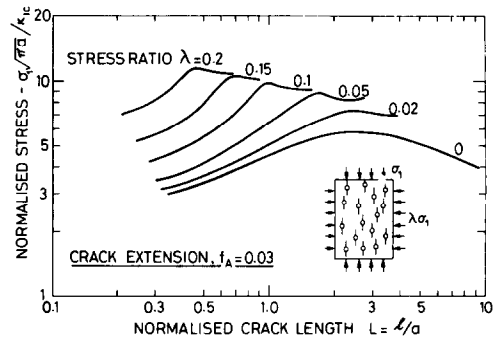


Fig. 22. A confining pressure limits crack growth, in the way shown schematically in Fig. 1, and quantitatively in this figure.

a peak which rises as the confining pressure is increased. More important, crack growth becomes more stable as confining pressure increases, and this, we believe, is the underlying cause of the transition from slabbing to shear-band formation and, finally, to homogeneous distributed microcracking, which was sketched in Fig. 1.

6. SUMMARY AND CONCLUSIONS

The growth of cracks from a cylindrical or spherical hole in a compressive stress field has been analysed. After an initial period of unstable growth, the cracks grow stably, meaning that the loads must be increased to cause further crack extension. If one principal stress is tensile, the cracks ultimately become unstable again and propagate catastrophically.

Experiments show that cracks interact with the surfaces of a finite sample in a way which causes them to grow faster than they do in an infinite medium. A simple analysis based on beam theory is developed to describe this interaction, and is extended to describe crack–crack interaction in a body containing many closely spaced cracks. This forms the basis for the development of a simple damage mechanics for brittle solids in compression. The method involves calculating the stress and modulus of the body for a given level of crack damage; from these the strain is obtained and stress–strain curves constructed. The influence of initial porosity and of stress state on the stress–strain curve is illustrated.

The behaviour of brittle porous solids in compression is very complicated. We have not, in this paper, arrived at a complete description of it. But the method and the results look promising: in particular, the growth of an isolated crack is explained; the approach to crack–surface interaction is in agreement with observations; and the method of handling crack-crack interaction in a large population of cracks avoids the immense mathematical complexity which attends other approaches to this problem; finally, the results so far indicate that the development of a "damage-mechanics" for brittle solids in compression has much potential.

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APPENDIX 1: CALCULATION OF STRESS-INTENSITY FACTORS FOR THE 2-D PROBLEM

Crack growth from a hole in an infinite plate

Jaeger and Cook [4] specialize the results given by Timoshenko and Goodier [3] for the stress field in the vicinity of a cylindrical hole of radius a when the principal stresses at infinity are σ_1 and $\sigma_3 = \lambda\sigma_1$ (positive when tensile). We define a new co-ordinate system (X, Z) which has its origin at the periphery of the hole along the x axis. Referring to Fig. 2, for $x \geq a$

$$X = x/a - 1$$
$$Z = z/a. \tag{A1}$$

In this co-ordinate system, the normal stress along the axial plane aligned with the principal stress axis is

$$\sigma_{zz}(X, 0) = -\sigma_1\left[\lambda + \frac{(1+\lambda)}{2}\left(\frac{1}{1+X}\right)^2 - \frac{3(1-\lambda)}{2}\left(\frac{1}{1+X}\right)^4\right]. \tag{A2}$$

Equation (2) has been plotted in Fig. 9, which shows that $\sigma_{zz}(X, 0)$ is tensile between $X = 0$ (where $\sigma_{zz} = \sigma_1$) and $X \simeq 0.7$.

The tensile stress generates a mode 1 stress intensity, K_I, at the tip of radial crack emanating from the hole. It has the form [5]

$$K_I = -F(\lambda, L)\sigma_1\sqrt{\pi l} \tag{A3}$$

where l is the crack-length (measured from the edge of the hole), $L = l/a$, σ_1 is the more compressive of the two principal stresses and $\lambda\sigma_1$ is the other principal stress. The function $F(\lambda, L)$ is tabulated in Sih [5] as a function of λ and L. The case of biaxial compression shown in Fig. 2 can be constructed from his tables† by superposition. The resulting values for $F(\lambda, L)$ are listed in Table 1 and plotted as full lines in Fig. 10.

Approximate stress intensities can be found from the Green's function for a crack embedded in the stress field of the hole [9]. Using the co-ordinate system defined in equation (A1), the stress intensity factor is

$$K_I = \frac{\sqrt{a}}{\sqrt{\pi L}}\int_{-L}^{+L}\sigma_{zz}(X, 0)\left(\frac{L+X}{L-X}\right)^{1/2} dX \tag{A4}$$

where $\sigma_{zz}(X, 0)$ is the normal stress acting across the crack plane, given by equation (A2), and $L = l/a$ as before. The result is approximate because the circle with emanating cracks of total normalised length $2(L + 1)$ is replaced by a single flat crack of length $2L$, making equation (A4) inaccurate when $L \ll 1$. Values of $F(\lambda, L) = -K_I/\sigma_1\sqrt{\pi l}$, calculated from equation (A4) are listed in Table 2 and plotted

†Note, however, that Sih [5] has tabulated $K_I = F(\lambda, L)\,\sigma\sqrt{\pi a}$ *not* $K_I = F(\lambda, L)\,\sigma\sqrt{a}$ as he claims, and that $\lambda = 0$ and $\lambda = 1$ columns have been switched. They are correctly plotted in Rooke and Cartwright [8].

Table 1. Stress intensity factors for two radial cracks emanating from a circular hole with biaxial compressive stresses given at infinity (see Fig. 2). From a superposition of solutions tabulated by Sih (1973)

	$K_1 = F(\lambda, l/a)\sqrt{\pi l}\,\sigma_1$					
	$\lambda = 0$		$\lambda = 0.1$		$\lambda = 0.2$	
l/a	F	$[\sqrt{l/a}\,F]^{-1}$	F	$[\sqrt{l/a}\,F]^{-1}$	F	$[\sqrt{l/a}\,F]^{-1}$
0.01	1.0743	9.308	0.7471	13.385	0.4199	23.820
0.02	1.0405	6.796	0.7181	9.846	0.3958	18.866
0.04	0.9683	5.164	0.6582	7.597	0.3481	14.365
0.06	0.9020	4.526	0.6034	6.766	0.3048	13.393
0.08	0.8421	4.198	0.5540	6.382	0.2658	13.302
0.10	0.7877	4.015	0.5091	6.211	0.2305	13.719
0.15	0.6712	3.847	0.4131	6.251	0.1549	16.664
0.20	0.5773	3.873	0.3360	6.654	0.0947	23.601
0.25	0.5002	3.998	0.2728	7.330	0.0455	43.974
0.30	0.4369	4.179	0.2213	8.249	0.0057	320.094
0.40	0.3394	4.659	0.1423	11.115	−0.0549	—
0.50	0.2688	5.261	0.0855	16.535		
0.60	0.2169	5.953	0.0442	29.183		
0.80	0.1472	7.598	−0.0103	—		
1.00	0.1047	9.555				
1.2	0.7691	11.870				
1.5	0.05125	15.931				
2.0	0.02866	24.673				
3.0	0.01129	44.279				

Table 2. Stress intensity factors for two radial cracks emanating from a circular hole with biaxial compressive stresses given at infinity (see Fig. 2). Approximate integral solutions discussed in text

	$K_1 = F(\lambda, l/a)\sqrt{\pi l}\,\sigma$					
	$\lambda = 0$		$\lambda = 0.1$		$\lambda = 0.2$	
	F	F	F	F	F	F
l/a	Eqn (4)	Eqn (5)	Eqn (4)	Eqn (5)	Eqn (4)	Eqn (5)
0.01	0.957	1.051	0.665	0.730	0.373	0.410
0.10	0.732	0.812	0.472	0.525	0.213	0.240
0.20	0.557	0.625	0.324	0.368	0.091	0.110
0.30	0.434	0.493	0.221	0.257	0.007	0.020
0.40	0.345		0.146			
0.50	0.279		0.092			
1.00	0.116					
1.50	0.059					

as the open circles in Fig. 10 where they are compared with the exact result. As would be expected, this approximation is good when L is large, poorer when L is small, though even here it is not bad.

When L is small, a better approximation is obtained by neglecting the curvature of the hole and approximating the stress intensity factor by that for an edge-crack in a half-plane. The Green's function [5] is

$$K_I = \frac{2\sqrt{\pi l}}{\pi}\int_0^L \sigma_{zz}(X,0)\left[\frac{1+G(X/L)}{(L^2-X^2)^{1/2}}\,dX\right] \qquad (A5)$$

where

$$G(X/L) = [1-(X/L)^2]\{0.29 - 0.39(X/L)^2 + 0.77(X/L)^4 - 0.99(X/L)^6 + 0.51(X/L)^8\}.$$

Numerical integration of equation (A5) yields the values of $F(\lambda, L) = -K_I/\sigma_1\sqrt{\pi a}$ given in Table 2 and plotted as the triangles in Fig. 10. At $L \lesssim 0.1$ where equation (A4) begins to fail, the edge-crack approximation (A5) is good.

Crack growth from a hole in a finite plate: interaction with surfaces

Consider a plate with height h and width w containing a central hole of radius a, and load with principal stresses σ_1, σ_3 (positive when tensile) as shown in Fig. 12. Cracks extend from the hole. If the dimensions of the plate are much larger than the length of the crack, the stress intensity at its tip is given by equation (3). But if the sample is narrow, the crack splits the plate into vertical beams or columns of length $2(l+a)$ and width $t = w/2$, each with a central segment where the section is reduced by the hole. This reduced section introduces bending moments into the beams. The line of action of the load F does not act along the neutral axis of the beam, due to its eccentricity (Fig. A1). Let the circular hole be replaced by an equivalent rectangular slot of height $2a$ and depth αa (where α is a constant, near unity†) and let h and g be the distances from the line of action of the load F to the neutral axes of the full and the reduced sections of the beam respectively. The moments acting on each section (assuming that deflections are small) are

$$m_1 = hF \quad \text{and} \quad m_2 = gF \qquad (A6)$$

where the suffix "1" refers to the full section, and "2" refers to the reduced section. Since these moments are constant, the gradient in each section is given by

$$\frac{dy}{dx} = \frac{mx}{EI} + \text{constant} \qquad (A7)$$

where x is measured along the beam. The gradient is zero at the ends and at the centre of the beam, and the gradients of the two sections are equal where they meet. Then (assuming that the second moments of area I_1 and I_2 are approximately equal)

$$h = \frac{\alpha^2 a^2}{2(l+a)} \quad g = \frac{-\alpha a l}{2(l+a)} \qquad (A8)$$

†We have found that a value $\alpha = 0.6$ gives the best correspondence between theory and experimental data.

and the moments are

$$m_1 = \frac{F\alpha^2 a^2}{2(l+a)} \quad m_2 = \frac{-F\alpha a l}{2(l+a)} \tag{A9}$$

These moments bend the columns as shown in Fig. A1. elastic energy is stored in the bent columns, and this energy changes if the crack advances, giving a further contribution to the crack driving force, G. The energy stored in one of the beams (shown in full lines in Fig. A1) is

$$U = \frac{1}{EI}\left\{m_1^2 l + m_2^2 a\right\} = \frac{F^2\alpha^3 a^3 l}{4(l+a)EI} \tag{A10}$$

and the strain energy release rate (or crack driving force) is

$$G = \frac{1}{b}\frac{dU}{dl} = \frac{F^2\alpha^4 a^4}{4EI(l+a)^2 b}. \tag{A11}$$

Relating this to the stress intensity by $K_I = \sqrt{EG}$ gives the extra contribution to the stress intensity due to bending

$$K_I^{B1} = -\left(\frac{3}{\pi}\right)^{1/2}\left(\frac{a}{t}\right)^{1/2}\frac{\alpha^2}{(L+1)}\sigma_1\sqrt{\pi a} \tag{A12}$$

where $L = l/a$. (The negative sign is lost in the energy method when the moments are squared; taking the square root we select the solution with the negative sign for obvious physical reasons.)

A confining stress, σ_3, applied a distributed load ($\sigma_3 b$ per unit length) to the outer surface of the beam of Fig. A1. This generates an additional moment m_3, tending to cancel m_1 and m_2 and of magnitude

$$m_3 = 2\sigma_3 b(l+a)^2\left\{\left(\frac{x}{2(l+a)}\right)^2 - \frac{x}{2(l+a)} + \frac{1}{6}\right\} \tag{A13}$$

where x is measured from one end of the beam. The energy in one beam of length $2(l+a)$, thickness t and depth b is

$$U_3 = \frac{1}{2EI}\int_0^{2(l+a)} m^2 dx = \frac{\sigma_3^2 b^2 (l+a)^5}{45EI}.$$

The crack driving force is

$$G = \frac{1}{b}\frac{dU}{dl} = \frac{4}{3}\frac{\sigma_3^2}{Et^3}(l+a)^4$$

giving a contribution to K_I of $\sqrt{EG}$, or

$$K_I^{B3} = +\left(\frac{4}{3\pi}\right)^{1/2}\left(\frac{a}{t}\right)^{3/2}(L+1)^2\sigma_3\sqrt{\pi a}. \tag{A14}$$

The total stress intensity due to bending is the sum of equation (A12) and (A14).

This is still not the whole story. We have assumed that the outward deflection of the beams was small and that changes of geometry could be ignored. But when the crack is long and the loads high, the large deflections increase the bending moment caused by σ_1; and at sufficiently high loads the beam buckles and deflections become very large indeed. We include this in an approximate but adequate way by noting that the beam will buckle at its Euler load

$$F_{crit} = \frac{\pi^2 EI}{(l+a)^2}$$

and that when it does, the stress intensity K_I becomes infinite. A perfectly straight, uniform, beam buckles suddenly and catastrophically at F_{crit}; but a beam which is already bent collapses progressively, so that the deflections become increasingly large as the end-load F approaches F_{crit}. We linearize this progressive collapse (it is the first order solution to the problem treated in this section, allowing for the change of geometry of the beam as it bends) by multiplying the bending deflection, and thus the stress intensity K_I, by the correction factor

$$f = \frac{1}{\left(1 - \dfrac{F}{F_{crit}}\right)} = \frac{1}{\left(1 + \dfrac{12}{\pi^2}\dfrac{(l+a)^2}{t^2}\dfrac{\sigma_1}{E}\right)}. \tag{A15}$$

Equation (6) of the text follows immediately.

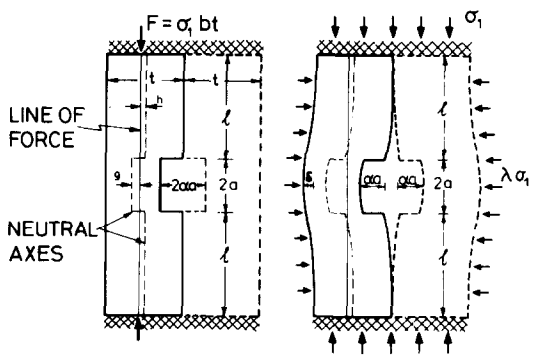


Fig. A1. The geometry of the cracked sample, showing the idealised, reduced, central section, the neutral axis and the line of force through each beam.

APPENDIX 2: CALCULATION OF THE STRESS-INTENSITY FACTORS FOR THE 3-D PROBLEM

Crack growth from a pore in an infinite body

We used the approximate method described in Appendix 1 to calculate K_I at the tips of cracks emanating from a spherical pore in a triaxial stress field.

The elastic stress state near a spherical hole with a uniaxial tension applied at infinity is given by Goodier [3]. We obtain the tensile stress $\sigma_{\theta\theta}$ at the pole (Fig. 16) in a triaxial compressive field by superposition of Goodier's solutions with the result

$$\sigma_{\theta\theta}(R,0) = \lambda\sigma_1 + \frac{\sigma_1}{2(7-5\nu)}\{12(\lambda-1)1/R^5 + [9 - 15\nu + 2\lambda(5\nu-1)]1/R^3\} \tag{B1}$$

where ν is Poisson's ratio, $R = r/a$ and $\lambda = \sigma_3/\sigma_1$ as before. This stress distribution is plotted in Fig. 9 for $\lambda = 0$, where it is compared with the equivalent two-dimensional result. The stress concentration at the sphere is only about one half as strong as that at a cylinder, and it falls more rapidly with distance (as r^{-5} compared to r^{-3}). So we expect cracks at a spherical hole to be shorter than those at a cylindrical hole, for the same stress.

The stress intensity K_I for a pair of flat cracks emanating from the poles of a spherical pore (Fig. 16) is found by using the approximate methods which were tested in Appendix 1. Then K_I is given by equation (A4) when the cracks are long and by equation (A5) when they are short, using equation (B1) for $\sigma_{\theta\theta}$ in place of σ_{zz} and replacing $X = x/a - 1$ by $r/a - 1$. In doing this we have approximated the integral over the face of the crack by an integral along the line of maximum tensile stress, but the stresses are so sharply peaked near the poles of the pore that other geometric considerations are of minor importance.

As in two dimensions, we write the stress intensity factor as

$$K_I = -F(\lambda, L)\sigma_1\sqrt{\pi l}.$$

The results of the numerical integration of equations (A4) and (A5), with $\nu = 0.25$, are given as $F(\lambda, l)$ in Table 3 and plotted in Fig. 17.

APPENDIX 3: STRESS INTENSITY CAUSED BY CRACK–CRACK INTERACTION

Consider the network of beams which describe the ligaments of solid between the array of cracks shown in Fig. 19. It is subjected to remote stresses σ_1 and σ_3 (positive when

Table 3. Stress intensity factors for two radial cracks emanating from a spherical pore with triaxial compressive stresses given at infinity (see Fig. 5). Approximate integral solutions as discussed in the text. Poisson's ratio is 0.25

	$K_1 = F(\lambda, l/a)\sqrt{\pi l}$							
	$\lambda = 0$		$\lambda = 0.05$		$\lambda = 0.1$		$\lambda = 0.2$	
l/a	F Eqn (3)	F Eqn (4)	F Eqn (3)	F Eqn (4)	F Eqn (3)	F Eqn (4)	F Eqn (3)	F Eqn (4)
0.01		0.611		0.500		0.388		0.165
0.05		0.523	0.378	0.418	0.283	0.313	0.091	0.103
0.10	0.391	0.436	0.302	0.337	0.212	0.238	0.033	0.041
0.20	0.275	0.311	0.194	0.222	0.113	0.133		
0.30	0.199		0.124		0.050			
0.40	0.148		0.078		0.008			
0.50	0.113		0.047					
0.60	0.089		0.024					
0.70	0.071		0.008					
0.80	0.057							
1.0	0.040							
1.5	0.020							

tensile). The beams have length l', thickness t' and depth b, and lie at the angle θ to the compression axis, such that, as the cracks become long

$$l' = \frac{l + a}{\cos\theta} \tag{C1}$$

$$t' = t\cos\theta \tag{C2}$$

$$\tan\theta = \frac{a}{l + a} \tag{C3}$$

and

$$(l + a)t = l't' = \frac{1}{2N_A} \tag{C4}$$

where t and $l + a$ (the dimensions of the unit cell) are defined in Fig. 19.

The network of beams is idealized in the third column of Fig. 19, and in Fig. A2 which shows the forces acting on each beam: an axial force T, a shear force S and a moment m. Simple geometry gives

$$T = \sigma_1 bt\cos\theta + \sigma_3 b(l + a)\sin\theta \tag{C5}$$

$$S = \sigma_1 bt\sin\theta - \sigma_3 b(l + a)\cos\theta \tag{C6}$$

and, for equilibrium, $m = \frac{1}{2}Sl'$. Both forces cause elastic energy to be stored in the beam: the first because it compresses the beam, the second because it bends it. It is easily shown that the elastic energy stored in the simple compressional field is, to first order, independent of crack length, so it does not contribute to the crack driving force. The shear S applied at either end of a beam of length l' and thickness t', built-in at both ends, stores energy $\frac{1}{2}S\delta$ (where δ is the end displacement), giving (via standard beam-theory) the elastic energy

$$U = \frac{S^2}{2E_0 b}\left(\frac{l'}{t'}\right)^3. \tag{C7}$$

Inserting equation (C6) for S and using the definitions of l' and t' [equations (C1) and (C2)] gives

$$U = \frac{\sigma_1^2 b}{2E_0}\frac{\tan^2\theta}{\cos^4\theta}\frac{(l + a)^3}{t}\left\{1 - \left(\frac{l + a}{t}\right)\frac{\lambda}{\tan\theta}\right\}^2. \tag{C8}$$

But t and $(l + a)$ are related to the number of holes per unit area N_A, by equation (C4); and the result is interesting only when the cracks are long (short cracks do not interact), when

$$\tan\theta \approx \frac{a}{l + a} \quad \cos\theta \approx 1$$

giving

$$U = \frac{\sigma_1^2 b}{E_0}a^2(l + a)^2 N_A\left\{1 - \frac{2(l + a)^3 N_A\lambda}{a}\right\}^2. \tag{C9}$$

The change in potential energy of the unit cell and its loads, ΔP, when the beams bend, is just $-U$.

We obtain the crack driving force, G, by differentiating P with respect to crack length l. Defining the area fraction of holes (the "porosity") as

$$f_A = \pi a^2 N_A$$

we obtain

$$G = -\frac{1}{b}\frac{dP}{dl} = \frac{2}{\pi}\frac{\sigma_1^2}{E_0}af_A(L + 1) \times\left\{\left[1 - \frac{8}{\pi}f_A\lambda(L + 1)^3\right]\left[1 - \frac{2}{\pi}f_A\lambda(L + 1)^3\right]\right\} \tag{C10}$$

where $L = l/a$ as before. The stress intensity K_1^I caused by crack–crack interaction is given by $K_1^I = \sqrt{E_0 G}$, giving

$$K_1^I = \frac{\sqrt{2}}{\pi}\sigma_1\sqrt{\pi a}f_A^{1/2}(L + 1)^{1/2} \left\{\left[1 - \frac{8}{\pi}f_A\lambda(L + 1)^3\right]\left[1 - \frac{2}{\pi}f_A\lambda(L + 1)^3\right]\right\}^{1/2}. \tag{C11}$$

The total stress intensity is the sum of this and that for an isolated crack in an infinite body, equation (3).

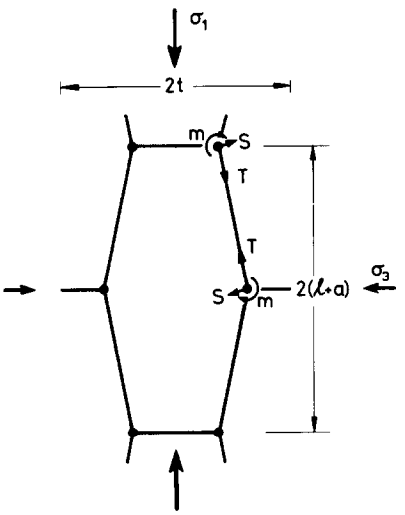


Fig. A2. The network of beams showing the forces and moments which act on each.